



UNIVERSIDAD DE ANTIOQUIA
1803

Effective two-component and two-mediator dark matter

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In collaboration with

Kimy Agudelo, Andrés Rivera and David Suárez (UdeA)



Baryon and Lepton Nonconserving Processes

Steven Weinberg (Harvard U.) (1979)

Published in: *Phys.Rev.Lett.* 43 (1979) 1566-1570

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#1 Systematic Study of One-Loop Dirac Neutrino Masses and Viable Dark Matter Candidates

Chang-Yuan Yao (Hefei, CUST), Gui-Jun Ding (Hefei, CUST) (Jul 31, 2017)

Published in: *Phys.Rev.D* 96 (2017) 9, 095004, *Phys.Rev.D* 98 (2018) 3, 039901 (erratum) • e-Print: [1707.09786](#) [hep-ph]

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Name	$SU(2)_L$	$U(1)_Y$
L	2	$-1/2$
H	2	$1/2$

$$\mathcal{L}_M = L \cdot H L \cdot H + \text{h.c}$$

Dark matter (radiative) realization $\rightarrow Z_2$

$$\mathcal{L}_D = (\nu_R)^\dagger L \cdot H + \text{h.c}$$

$$\overline{M_{N^c R^c R}} \rightarrow U(1)$$

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Scotogenic one-loop realizations

Majorana

Dirac

Systematic study of the d=5 Weinberg operator at one-loop order

#1

Florian Bonnet (Wurzburg U.), Martin Hirsch (Valencia U., IFIC), Toshihiko Ota (Munich, Max Planck Inst.), Walter Winter (Wurzburg U.) (Apr, 2012)

Published in: *JHEP* 07 (2012) 153 • e-Print: [1204.5862](#) [hep-ph]

pdf DOI cite claim

reference search 197 citations

Models with radiative neutrino masses and viable dark matter candidates

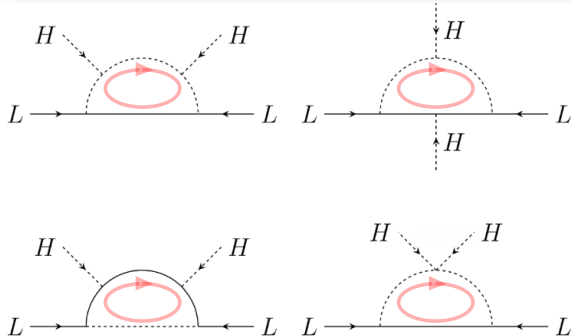
#1

Diego Restrepo (Antioquia U.), Oscar Zapata (Antioquia U.), Carlos E. Yaguna (Munster U., ITP) (Aug 16, 2013)

Published in: *JHEP* 11 (2013) 011 • e-Print: [1308.3655](#) [hep-ph]

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Pathways to Naturally Small Dirac Neutrino Masses

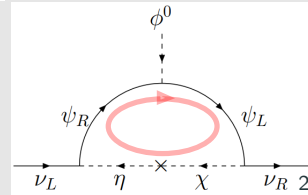
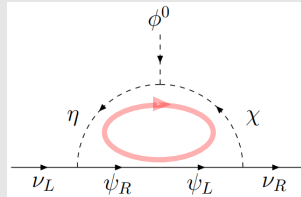
#1

Ernest Ma (UC, Riverside), Oleg Popov (UC, Riverside) (Sep 8, 2016)

Published in: *Phys.Lett.B* 764 (2017) 142-144 • e-Print: [1609.02538](#) [hep-ph]

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Scotogenic one-loop realizations

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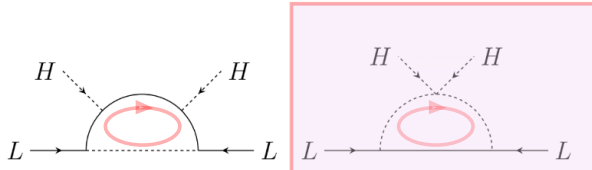
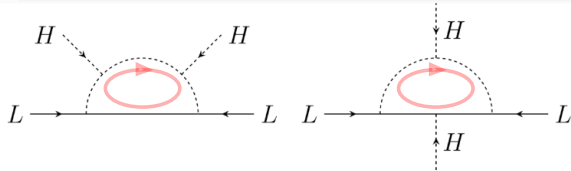
Models with radiative neutrino masses and viable dark matter candidates

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#1

Pathways to Naturally Small Dirac Neutrino Masses

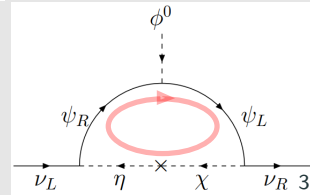
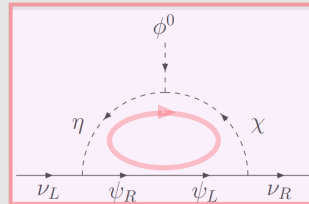
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Published in: *Phys.Lett.B* 764 (2017) 142-144 • e-Print: [1609.02538](#) [hep-ph]

pdf DOI cite claim

reference search 125 citations

#1



Radiative seesaw mechanism at weak scale

#1

Zhi-jian Tao (Beijing, Inst. High Energy Phys.) (Mar, 1996)

Published in: *Phys.Rev.D* 54 (1996) 5693-5697 • e-Print: [hep-ph/9603309](https://arxiv.org/abs/hep-ph/9603309) [hep-ph]

pdf DOI cite claim reference search 34 citations

Verifiable radiative seesaw mechanism of neutrino mass and dark matter

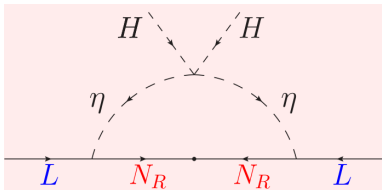
#2

Ernest Ma (UC, Riverside) (Jan, 2006)

Published in: *Phys.Rev.D* 73 (2006) 077301 • e-Print: [hep-ph/0601225](https://arxiv.org/abs/hep-ph/0601225) [hep-ph]

pdf DOI cite claim reference search 1,539 citations

Name	$SU(2)_L$	$U(1)_Y$	Z_2
N_R	1	0	—
η	2	1/2	—



Radiative Neutrino Mass, Dark Matter and Leptogenesis

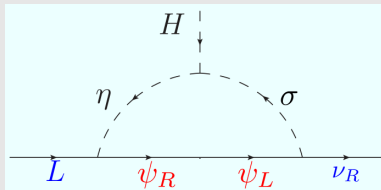
#1

Pei-Hong Gu (ICTP, Trieste), Utpal Sarkar (Ahmedabad, Phys. Res. Lab) (Dec, 2007)

Published in: *Phys.Rev.D* 77 (2008) 105031 • e-Print: [0712.2933](https://arxiv.org/abs/0712.2933) [hep-ph]

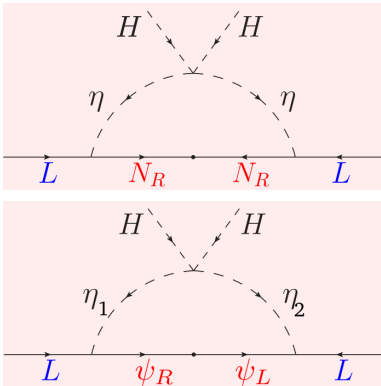
pdf DOI cite claim reference search 124 citations

Name	$SU(2)_L$	$U(1)_Y$	Z_2	$U(1)_L$
ν_R	1	0	+	-1
ψ_R	1	0	—	-1
ψ_L	1	0	—	-1
η	2	1/2	—	0
σ	1	0	—	0



Radiative seesaw mechanism at weak scale

Zhi-jian Tao (Beijing, Inst. High Energy Phys.) (Mar, 1996)

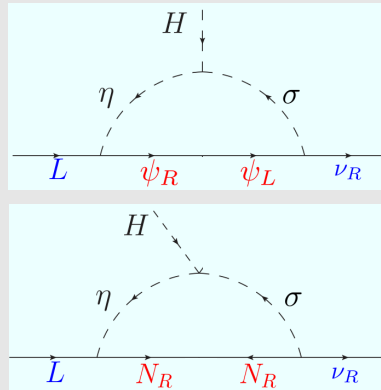
Published in: *Phys.Rev.D* 54 (1996) 5693-5697 • e-Print: [hep-ph/9603309](#) [hep-ph]
[pdf](#) [DOI](#) [cite](#) [claim](#) [reference search](#) [34 citations](#)
New Scotogenic Model of Neutrino Mass with $U(1)_D$ Gauge Interaction

Ernest Ma (UC, Riverside), Ivica Picek (Zagreb U., Phys. Dept.), Branimir Radovčić (Zagreb U., Phys. Dept.) (Aug 24, 2013)

Published in: *Phys.Lett.B* 726 (2013) 744-746 • e-Print: [1308.5313](#) [hep-ph]
[pdf](#) [DOI](#) [cite](#) [claim](#) [reference search](#) [86 citations](#)

Radiative Neutrino Mass, Dark Matter and Leptogenesis

Pei-Hong Gu (ICTP, Trieste), Utpal Sarkar (Ahmedabad, Phys. Res. Lab) (Dec, 2007)

Published in: *Phys.Rev.D* 77 (2008) 105031 • e-Print: [0712.2933](#) [hep-ph]
[pdf](#) [DOI](#) [cite](#) [claim](#) [reference search](#) [124 citations](#)


Dirac neutrino mass generation from a Majorana messenger

Julian Calle (Antioquia U.), Diego Restrepo (Antioquia U. and IIP, Brazil), Óscar Zapata (Antioquia U.) (Sep 20, 2019)

Published in: *Phys.Rev.D* 101 (2020) 3, 035004 • e-Print: [1909.09574](#) [hep-ph]
[pdf](#) [links](#) [DOI](#) [cite](#) [claim](#) [reference search](#) [15 citations](#)

Dark matter stability and Dirac neutrinos using only Standard Model symmetries

Cesar Bonilla (Munich, Tech. U.), Salvador Centelles-Chuliá (Valencia U., IFIC), Ricardo Cepedello (Valencia U., IFIC), Eduardo Peinado (Mexico U.), Rahul Srivastava (Valencia U., IFIC) (Dec 4, 2018)

Published in: *Phys.Rev.D* 101 (2020) 3, 033011 • e-Print: 1812.01599 [hep-ph]

Minimal radiative Dirac neutrino mass models

Julian Calle (Antioquia U.), Diego Restrepo (Antioquia U.), Carlos E. Yaguna (UPTC, Tunja), Óscar Zapata (Antioquia U.) (Dec 13, 2018)

Published in: *Phys.Rev.D* 99 (2019) 7, 075008 • e-Print: 1812.05523 [hep-ph]

pdf DOI cite claim reference search 46 citations

$$\mathcal{L}_5 = \frac{h}{\Lambda} (\nu_R)^\dagger L \cdot HS^* + \text{H.c.},$$

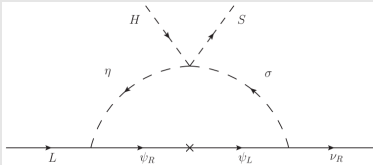


FIG. 6. T3-1-A-I ($\alpha = 0$): $B - L$ flux in the Dirac radiative seesaw.

	$(\nu_{Ri})^\dagger$	$(\nu_{Rj})^\dagger$	$(\nu_{Rk})^\dagger$	$(\psi_R)^\dagger \quad \psi_L \quad \eta_a \quad \sigma_a \quad \text{---}$					S
T3-1-A-I									
L	+4	+4	-5	r	$-r$	$1-r$	$4-r$	---	3

Anomaly cancellation of a U(1) gauge symmetry with chiral fermions

In a fundamental theory the elementary fermions are expected to be chiral, i.e., their charges should not allow a mass term larger than the scale of spontaneous symmetry breaking. For any set of charges associated to a U(1) gauge symmetry

$$\mathbf{Z} = [Z_1, Z_2, \dots, Z_N] ,$$

The triangle anomaly with three U(1) gauge bosons on the external lines, i.e., the $[U(1)]^3$ anomaly, and with one U(1) gauge boson and two gravitons on the external lines should also be cancelled

$$\sum_{\alpha=1}^N Z_{\alpha} = 0, \qquad \sum_{\alpha=1}^N Z_{\alpha}^3 = 0, \qquad (1)$$

$U(1)_Y$: no vector-like, i.e., pairs of fields with ‘equal-but-opposite’ charges

The hypercharge for one generation in the standard model can be seen as gauge Abelian symmetry $U(1)_Y$ with 15 left-handed Weyl massless fermions with integer charges

$$Y = [1, 1, 1, 1, 1, 1, -4, -4, -4, 2, 2, 2, -3, -3, 6]$$

which satisfy

$$\sum_i Y_i = 0, \quad \sum_i Y_i^3 = 0.$$

If we introduce an scalar, H , of charge 3 we can form Yukawa couplings through the Dirac pairs

$$u_\alpha = (1, -4), (1, -4), (1, -4), \quad d_\alpha = (1, 2), (1, 2), (1, 2), \quad e = (-3, 6)$$

and the *chiral fermions* acquire Dirac masses after the EWSB. They are degenerate because the additional color symmetry. One remains massless, $\nu_L = (-3)$. The remnant Z_3 symmetry guarantees the stability of the lightest quark

Secluded gauge $U(1)_D$ without vector-like fermions:

$$\mathbf{S} = [\chi_1, \chi_2, \dots, \psi_1, \psi_2, \dots, \psi_{N'}]$$

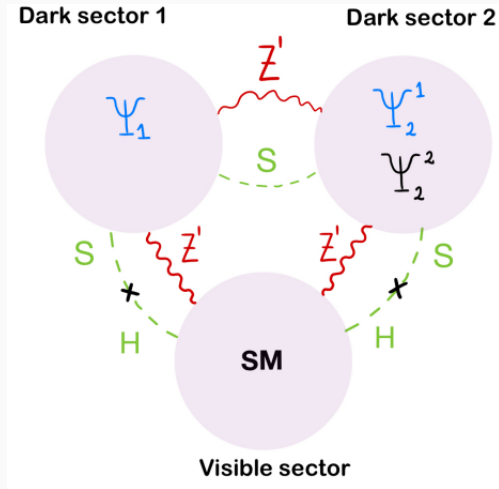
- *Dark Higgs mechanism*: Singlet dark Higgs ϕ acquires a vev and give mass to the dark photon

$$\mathcal{L} = i\psi_a^\dagger \overline{\sigma}^\mu \left(\partial_\mu - ig_D Z_\mu^D \right) \psi_a - \frac{1}{4} V_{\mu\nu} V^{\mu\nu} + \sum_{a < b} h_{ab} \psi_a \psi_b \phi^{(*)} + \text{h.c.} - V(\phi). \quad (2)$$

- S_α are the charges of SM-singlet right-handed chiral fermions with $N \geq 5$
 - χ_i *massless fermions* with $i = 1, \dots, N'$ with $N' \leq N$
 - ψ_a *multi-component dark matter*: massive after the spontaneous symmetry breaking of $U(1)_D$ with $a = N' + 1, \dots, N$
- *Larger parameter space*: Dark photon, Z_D , exclusions instead of Z'
- Z_D and ϕ masses are related by (arXiv:1610.03063)

$$h/g_D = \sqrt{2} m_\psi / m_{Z_D}.$$

Multi-component and two-mediator DM with kinetic mixing



Multi-component and two-mediator DM with kinetic mixing

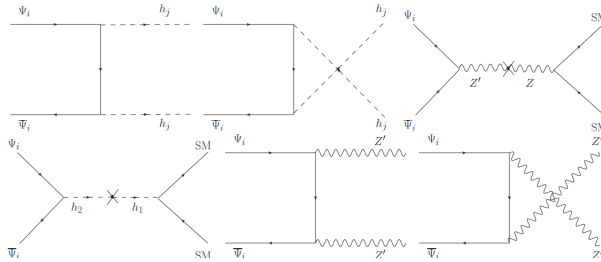


Figure 3: DM annihilation channels.

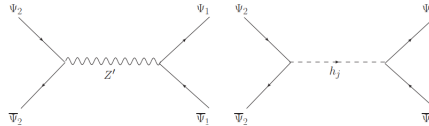


Figure 4: DM conversion channels $\Psi_1 \leftrightarrow \Psi_2$.

Decrease the number of charges to be assigned to dark matter particles, ψ_i below

$$[\chi_1, \chi_2, \dots, \psi_1, \psi_2, \dots, \psi_N]$$

Secluded case:

$$[\nu, \nu, (\nu), \psi_1, \psi_2, \dots, \psi_N]$$

$$\chi_1 \rightarrow \nu_{R1}, \dots, \chi_{N_\nu} \rightarrow \nu_{RN_\nu}, \quad 2 \leq N_\nu \leq 3,$$

$$\mathcal{L}_{\text{eff}} = h_{\nu}^{ij} (\nu_{Ri})^{\dagger} \epsilon_{ab} L_j^a H^b \left(\frac{\phi^*}{\Lambda} \right)^{\delta} + \text{H.c.}, \quad \text{with } i, j = 1, 2, 3,$$

ϕ is the complex singlet scalar responsible for the SSB of the anomaly-free gauge symmetry and give mass to all ψ_a

$$\phi = -\frac{\nu}{\delta},$$

Decrease the number of charges to be assigned to dark matter particles, ψ_i below

$$[\chi_1, \chi_2, \dots, \psi_1, \psi_2, \dots, \psi_N]$$

Secluded case:

$$[5, 5, -3, -2, 1, -6]$$

$$\chi_1 \rightarrow \nu_{R1}, \chi_2 \rightarrow \nu_{R2}, \quad N_\nu = 2,$$

$$\mathcal{L}_{\text{eff}} = h_\nu^{aj} (\nu_{Ra})^\dagger \epsilon_{bc} L_j^b H^c \left(\frac{\phi^*}{\Lambda} \right) + \text{H.c.}, \quad \text{with } j = 1, 2, 3,$$

Decrease the number of charges to be assigned to dark matter particles, ψ_i below

$$[\chi_1, \chi_2, \dots, \psi_1, \psi_2, \dots, \psi_N]$$

Secluded case:

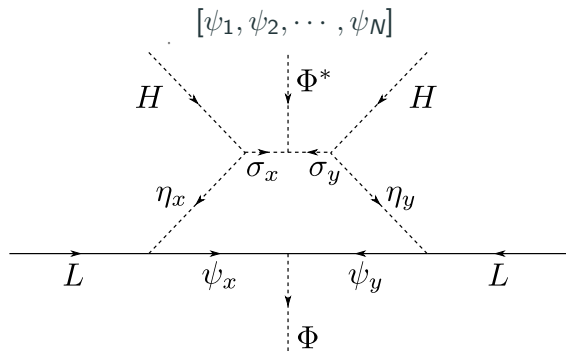
$$[5, 5, -3, -2, 1, -6]$$

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$$\mathcal{L}_{\text{eff}} = h_\nu^{aj} (\nu_{Ra})^\dagger \epsilon_{bc} L_j^b H^c \left(\frac{\phi^*}{\Lambda} \right) + \text{H.c.}, \quad \text{with } j = 1, 2, 3,$$

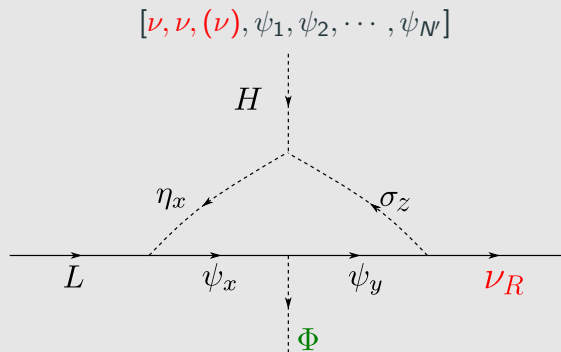
$$z = [5, 5, -3, -2, 1, -6] \rightarrow \phi = -5 \rightarrow [(5, 5), (-3, -2), (1, -6)]$$

$$\mathcal{L} \subset h_{(-3, -2)} \psi_{-3} \psi_{-2} \phi^* + h_{(1, -5)} \psi_1 \psi_{-6} \phi^* + \text{h.c.}$$



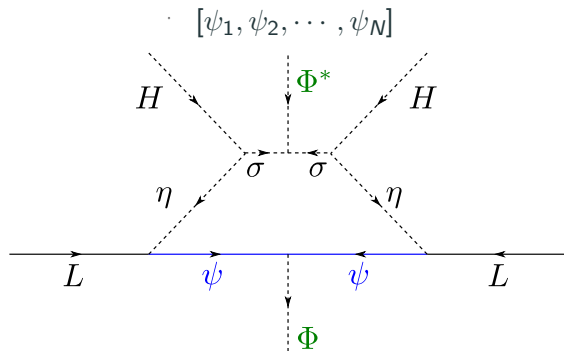
$$\frac{y}{\Lambda} \bar{L} \bar{L} H H \rightarrow \frac{y}{\Lambda} \bar{L} \bar{L} H H \frac{\phi}{\Lambda} \frac{\phi^*}{\Lambda}$$

ϕ give mass to all ψ_a

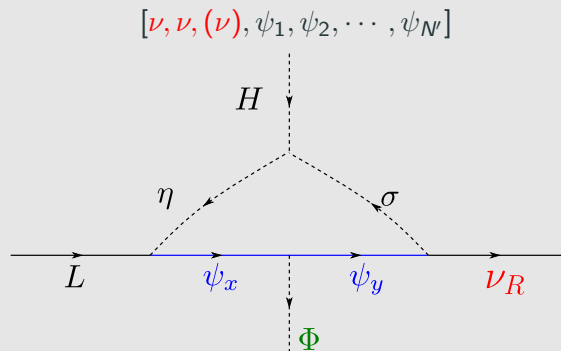


$$y(\nu_R)^\dagger \bar{L} H \rightarrow y(\nu_R)^\dagger \bar{L} H \frac{\phi^*}{\Lambda}$$

ϕ give mass to all ψ_a



$$\phi = 2 \rightarrow [\underbrace{1, 1}_{\psi_a}, (2, -4), (4, -6), (3, -5)]$$



$$\phi = 9 \rightarrow [[\underbrace{9, 9, 9}_{\psi_a}, (1, -10), (1, -10), (-4, -5)],$$

Anomaly-free chiral $U(1)_D$ and its scotogenic implication

Chi-Fong Wong (Macau U. Sci. Tech., SSI) (Aug 19, 2020)

Published in: *Phys.Dark Univ.* 32 (2021) 100818 • e-Print: [2008.08573](https://arxiv.org/abs/2008.08573) [hep-ph]

#1

Anomaly-free Abelian gauge symmetries with Dirac scotogenic models

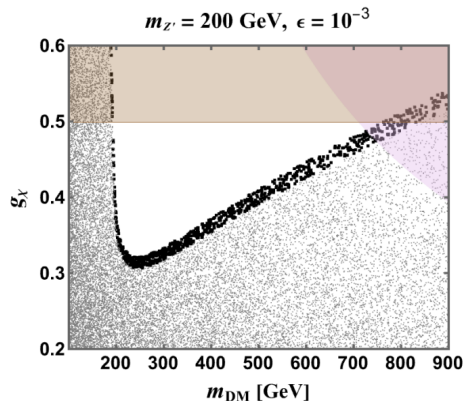
Nicolás Bernal (Antonio Narino U.), Julián Calle (Antioquia U.), Diego Restrepo (Antioquia U.) (Feb 11, 2021)

Published in: *Phys.Rev.D* 103 (2021) 9, 095032 • e-Print: [2102.06211](https://arxiv.org/abs/2102.06211) [hep-ph]

#8

$$\phi = 2 \rightarrow [1, 1, (2, -4), (4, -6), (3, -5)]$$

Secluded DM with Majorana fermion candidate



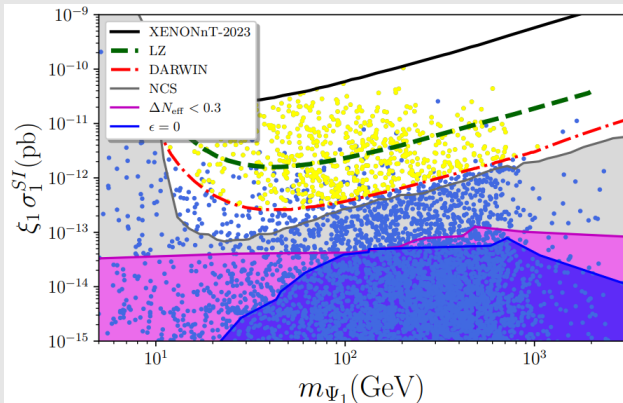
Chiral dark matter and radiative neutrino masses from gauged U(1) symmetry

K.S. Babu (Oklahoma State U.), Shreyashi Chakdar (Holy Cross Coll. and Munster U., ITP), Vishnu P. K (Sep 13, 2024)

e-Print: 2409.09008 [hep-ph]

$$\phi = 9 \rightarrow [[9, 9, 9, (1, -10), (1, -10), (-4, -5)],$$

New decay channel: $Z_D \rightarrow \bar{\nu}\nu$



In progress...